An Empirical Study of the Unique Character of Involvement in Multi-site STEM Education Evaluations

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As the number of large multi-site federal education programs increases, so too does the need for a more complete understanding of how to conduct evaluations of such programs. Lawrenz and Huffman (2003) suggest that given the scope and complexity of projects in multi-site program evaluations, traditional notions of participatory research, where stakeholders are involved to some degree in decision making and control, are simply not feasible. The objective of this study is to determine how multi-site evaluations have dealt with the challenges of effectively involving stakeholders in the face of increased complexity. Armed with this knowledge, educators and evaluators can better understand the nature of participatory evaluation in multi-site evaluations of education programs and, thus, better inform the development of evaluation policy and procedures.

Two questions guided this research:

(1) What patterns of evaluation involvement exist in large, multi-site evaluations?

(2) How does project- or site-level involvement in multi-site evaluations differ from individual stakeholder participation documented in single-site research?

In general, multi-site program evaluations differ from single-site evaluations in two main aspects: (1) multi-site evaluations include more than one site or project and (2) the evaluation may include cross-site evaluation activity or analysis (Straw & Herrell, 2002). Leff and Mulkern (2002) propose two grounding principles for multi-site evaluations in the social services: (1) the interventions should be science-based (i.e., use scientific investigation techniques) and (2) a broad range of stakeholders should participate. To expand the understanding of the ways in which participatory evaluation practices can be applied to large, multi-site settings and to explore

aspects of the use and influence of these evaluations, the National Science Foundation (NSF) awarded a University of Minnesota research team a multi-year grant, "Beyond Evaluation Use."

The primary population for this research on involvement is comprised of the principal investigators (PIs), project personnel, and evaluators from three NSF programs: (1) Advanced Technological Education (ATE), (2) Collaboratives for Excellence in Teacher Preparation (CETP), and (3) Local Systemic Change through Teacher Enhancement (LSC).[1] The three programs were similarly structured with distinct national or program-level evaluations and local project- or site-level participation either invited or required. In the multi-site context, the term "program" refers to the over-arching national program, which provides guidelines and funding for the smaller site-level "projects."

The research team recognized that the existing notion of participatory evaluation—where an evaluator and stakeholder team works together to conduct an evaluation—did not adequately capture participatory evaluation in large multi-site settings. Instead, the more inclusive construct of "involvement" was introduced to incorporate the full range of activity by any individual or group in any phase of an evaluation (planning it, conducting it, or doing something with the findings) and at any level of intensity (from attending a single meeting to actively engaging in all evaluation activities).

Participants in traditional participatory evaluations are generally individual stakeholders, e.g., recipients of services, staff, or community members. By contrast, participants in the large, multi-site participatory program evaluations are either individuals or groups affiliated with projects or sites (project PIs, project personnel, and project evaluators). This project-centric

[1] The Beyond Evaluation Use grant research also includes one other NSF program: Building Evaluation Capacity of Science, Technology, Engineering and Mathematics (STEM) Projects - Math Science Partnership Research Education (MSP-RETA). However, MSP-RETA did not conduct a distinct program level evaluation. Rather, as a project, the Utah State MSP- RETA promoted and assisted with evaluation for all other MSP projects. Consequently, there was no program level evaluation (at the time of our survey) to be directly involved with, so that program is not included in this involvement study.

identification of stakeholders ultimately results in the emergence of a single, collective stakeholder–the project–in large, multi-site evaluations (Lawrenz & Huffman, 2003).

Theoretical framework

Project involvement, as defined by Lawrenz and Huffman (2003), has two theoretical underpinnings: Cousins and Whitmore's (1998) systematic collaborative inquiry and Burke's (1998) key decision points, which delineate involvement across the spectrum of evaluation stages, activities, and levels of control. Lawrenz and Huffman (2003) applied these two theoretical frameworks to five NSF-funded multi-site evaluations and concluded that project involvement varied on four types of decision making: types of information to collect, whether or not to participate, what data to provide, and how to use the information. They proposed a continuum of participation in decision-making in multi-site STEM program evaluations:

- One extreme offers few opportunities for local participation in decision-making, in that the program evaluation is conducted by an outside entity that collects the data about the projects and makes decisions to address the needs of funders;
- In the middle, project participation is mandated in that each project is required to collect its own data according to standard procedures and to submit it to a central external evaluator; and
- The other extreme offers high levels of project involvement in decision-making, in that projects independently design and conduct their own evaluations according to their own needs.

The three programs in this study fall across the continuum, from ATE with few opportunities for participation, to LSC in the middle with mandated project participation, and

finally to CETP with high project involvement opportunities with individual projects encouraged, but not required, to participate.

Method

Using the cross-case study methodology of Yin (2003) and Merriam (1998), the research team with input from expert advisors developed in-depth, analytic case descriptions of the participants' levels of involvement in the three evaluation studies. The case studies identified the extent to which projects were involved in the program-level evaluations. Data for the case studies were collected through an online survey distributed to project leaders and evaluators, interviews with selected project leaders and evaluators, and interviews with the respective NSF program officers. Additional data were collected through archival document review, publications, and reports produced by the programs and their respective program evaluations and through reflective descriptions provided by the program evaluation PIs. This study on involvement relied primarily on data from online survey and project interviews, both described in detail below. In addition, a literature review of participatory evaluation research allowed for a systematic comparison of the ways in which researchers and evaluators have documented stakeholder involvement and what participants in this research on multi-site evaluations reported on the survey and during interviews.

Survey. Data were collected through an online survey distributed to project leaders and evaluators. The final survey population–the number of ATE, CETP, and LSC PIs and evaluators with valid email addresses–was 645, with an overall response rate of 46% (313/645). Among the respondents, 80% identified themselves as PIs or project staff and 20% as evaluators.

Using Burke's Key Decision Points as a general framework, the involvement scale measured depth of participation by the extent to which stakeholders took part in the key

decisions. The EUG team assigned each of Burke's key decisions to one of three evaluation stages: evaluation planning, implementation, and communication of results.

Involvement items included the following, "I was involved in…"

- The discussions that focused the evaluation.
- Identifying evaluation planning team members.
- Developing the evaluation plan.
- Developing data collection instruments.
- Developing data collection processes.
- Collecting data.
- Reviewing collected data for accuracy and/or completeness.
- Analyzing data.
- Interpreting collected data.
- Writing evaluation reports.
- Reviewing evaluation reports for accuracy and/or completeness.
- Presenting evaluation findings (e.g., to staff, to stakeholders, to an external audience).
- Developing future project plans based on evaluation results.

The response options for each item were: (1) No, (2) Yes-a little, (3) Yes-some, (4) Yes-extensively, or (5) I don't think this activity took place. The involvement scale had a high reliability (Cronbach's Alpha = .943) and was validated for use in assessing involvement levels in multi-site STEM evaluations in September 2007 (Toal, 2007). A non-respondent survey found that non-respondents did not have significantly different levels of involvement from respondents.

Project Interviews. To date, follow-up interviews have been conducted with 12 survey respondents who indicated their willingness to be interviewed. The interviews covered their motivation for involvement, perceived influence on the evaluation, and the ways in which the evaluation was used or had influence. Relevant questions included:

- Will you please describe the ways in which you were involved in the program evaluation?

- Overall, how would you rate the level of your involvement in the program evaluation? Would you say that you were not at all involved, involved a little, involved some, or involved extensively?

Results

The first research question addressed patterns of involvement in multi-site evaluations of large education programs. A descriptive analysis of the survey items showed that all tasks had a majority of respondents performing them at least to a little extent. Items appear to have a bimodal distribution, suggesting that generally respondents were typically either not at all involved or extensively involved in the activity. Of the thirteen involvement items, the highest levels of extensive involvement among respondents were (a) involvement in discussions that focused the evaluation and (b) reviewing evaluation reports for accuracy and completeness. The items with the highest rates of no involvement whatsoever were (a) identifying evaluation planning team members, (b) writing evaluation reports, and (c) analyzing data.

Interviewees described specific activities with which they were involved, providing a clearer picture of which activities fostered feelings of involvement. Some examples of the ways in which interviewees reported being involved with these program evaluations include: planning the evaluation, helping write survey questions, providing feedback on instruments, completing online surveys, collecting data, attending conferences put on by the evaluation team, and attending training on data collection, among others.

A systematic comparison of the interview and survey responses suggests a disconnect between the number of activities and overall perceptions of involvement, though there is not a precise way to compare scale and interview responses. The evidence suggests, however, that individuals perceive involvement differently. Indeed, the case studies revealed that intense or burdensome involvement in even just one activity can result in people feeling as if they were

deeply involved. Therefore, increasing the number of activities does not always translate into stronger feelings of involvement.

The second research question addressed differences of project involvement in multi-site evaluations from the more traditional notions of stakeholder participation, as documented in previous research. To address this question the team conducted an extensive review of empirical research on participatory evaluation. Comparing those findings to the ways in which our study participants reported being involved allowed us to highlight the differences between our findings and previous research. The findings of this empirical literature review are summarized here, but the detailed findings are outside the scope of this paper and are reported elsewhere.

There is considerable overlap in these findings and the findings in previous empirical literature, but there are some notable differences of interest. First, although the scale did not include an involvement item related to attendance at annual meetings or conferences, nor is it mentioned in the literature, interviewees often cited such attendance as involvement. In addition, in contrast to the literature, "developing future project plans based on evaluation results," was also seen as a way of involvement.

Only one item cited in the empirical research, "writing recommendations" was not directly supported by this current study. However, that task may be indirectly addressed through the "writing the evaluation report" scale item. Because recommendations are always a standard part of evaluation reports, the "writing recommendations" could reasonably be included as part of the "writing final report." Brandon's 1999 study provides an example to support this assertion that the two tasks are often interwoven. The evaluators in the Brandon (1999) article used student forums to solicit feedback of drafts of the recommendations that were to be included the evaluation's annual reports. So, although final reports do not always include recommendations, it makes sense that recommendations are presented in report form and, therefore, can be subsumed

under the writing report scale item, an activity that was documented in the survey, but not the interviews.

A final key difference is the impact of completing a survey on perceptions of involvement. Traditionally, evaluators have not considered providing data (either in a survey or in an interview) to be stakeholder participation, with only one exception (Earl, 1995). However, interviewees, especially in the ATE with its annual online surveys, consistently cited this task as a way in which they were involved.

Conclusions

The results of the case studies suggest that despite similarities among the evaluations, patterns of involvement varied within each of the three evaluations. For example, each program evaluator hoped to involve the sites in some aspects of the planning, all required some sort of data collection, and all produced overall reports of the findings to share with the sites. However, perceptions of involvement from the sites varied according to both the breadth and depth of the activities, as well as to how those activities fostered feelings of involvement.

Additionally, the systematic comparison of the participatory evaluation literature and our research findings suggest involvement in multi-site evaluations is different from traditional conceptions of participatory research documented in empirical research. First, attendance at meetings and conferences, even if the meetings were not directly related to the program evaluation, created feelings of involvement among the projects. Conference or meeting attendance has not been previously documented as a mode of participation in any of the participatory literature. Second, although our research indicated that projects were involved with planning for use, none of the empirical research discussed this, at least not under the umbrella of participation.

Finally, our findings suggest that completing mandatory data requests or evaluation surveys were also cited as involvement. When asked about the ways in which they were involved in the program evaluation, the majority of interviewees cited some form of data collection. This finding differs from the empirical research in that it is a much more passive form of involvement than has been documented thus far.

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